

Flight Block Time Forecasting and Schedule Reliability Optimization

Part 1 Report: Data Cleaning, Feature Engineering, Exploratory Analysis, and Initial Schedule Buffer Recommendations

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Data period: Oct 01, 2025 to Dec 31, 2025

Dataset: BTS On-Time Reporting data for American Airlines flights connected to Virginia/DC-area airports

Project phase	Part 1 - cleaned dataset and exploratory analysis
Carrier analyzed	American Airlines (AA)
Geographic scope	Flights with one endpoint in the Virginia/DC-area airport set: DCA, IAD, RIC, ORF, ROA, CHO, PHF
Final cleaned records	17,748 completed AA flights
Routes analyzed	91 origin-destination routes
Main outcome variables	Block time error and late arrival indicator

1. Summary of the First Project Phase

This first phase builds the analytical foundation for a flight block time forecasting and schedule reliability project. The goal is to understand how scheduled flight block times compare with actual gate-to-gate elapsed times, where reliability issues appear most often, and whether those issues look related to schedule block-time assumptions or broader operational factors.

The analysis used October, November, and December 2025 BTS On-Time Reporting data. After combining the monthly files, the dataset was cleaned to retain completed, non-diverted American Airlines flights with complete values for scheduled elapsed time, actual elapsed time, delay fields, taxi times, and distance. The final cleaned dataset contains 17,748 AA flights across 91 routes.

- Overall late-arrival rate for the cleaned AA dataset was 24.2%.
- The average scheduled block time was 145.2 minutes, while the average actual elapsed time was 140.4 minutes.
- The average block time error was -4.8 minutes, meaning the actual elapsed time was lower than the scheduled elapsed time on average.
- About 28.2% of flights were underblocked at the individual-flight level, meaning actual elapsed time exceeded scheduled elapsed time.
- Route-level results show that late arrivals are not always caused by insufficient scheduled block time. Several routes had high late-arrival rates even when average actual block time was lower than scheduled block time.
- The initial recommendation table separates routes into categories: review block-time buffers, investigate operational root causes, potentially reduce excess block time, or monitor.

2. Project Objective

The broader project objective is to forecast actual flight block time and use those forecasts to support schedule reliability decisions. Block time is important because it directly affects both reliability and operational efficiency.

- If scheduled block time is too low, flights may arrive late even when operations are otherwise normal.
- If scheduled block time is too high, the schedule may become inefficient because aircraft and crews spend more planned time than necessary.
- The practical goal is to find a balance between reliability and efficient scheduled minutes.

The first part of the project does not yet build a predictive model. Instead, it prepares the dataset, defines the operational metrics, and identifies route and departure-time patterns that should guide the modeling phase.

3. Data Scope and Preparation

3.1 Data Source and Scope

The input data consisted of three monthly BTS On-Time Reporting CSV files for October, November, and December 2025. The analysis was scoped to American Airlines flights connected to Virginia/DC-area

airports. The final cleaned dataset includes only completed flights, because cancelled and diverted flights do not provide valid actual elapsed time for block-time comparison.

Date range	Oct 01, 2025 to Dec 31, 2025
Months included	October, November, December 2025
Carrier	American Airlines (AA)
Cleaned completed flights	17,748
Distinct routes	91
Average distance	799.1 miles
Average scheduled block time	145.2 minutes
Average actual elapsed time	140.4 minutes

3.2 Cleaning Steps

1. Loaded October, November, and December CSV files and concatenated them into one combined dataset.
2. Standardized column names to lowercase for easier analysis.
3. Filtered out cancelled and diverted flights because actual elapsed time is not comparable for those records.
4. Dropped rows with missing values in the key fields used for block-time and reliability analysis.
5. Filtered the cleaned dataset to American Airlines flights.
6. Created route, departure-hour, block-time error, underblocked flag, late-arrival flag, and weekend indicator features.

4. Feature Engineering and Metric Definitions

The following features were created to support route-level and time-of-day reliability analysis.

Feature	Definition
route	Origin and destination pair, such as DCA-BOS or CLT-DCA.
dep_hour	Scheduled departure hour extracted from scheduled departure time.
block_time_error	Actual elapsed time minus scheduled elapsed time. Positive values indicate the flight took longer than scheduled.
underblocked	Indicator equal to 1 when block_time_error is positive.
late_arrival	Indicator equal to 1 when arrival delay is greater than 15 minutes.
is_weekend	Indicator equal to 1 for Saturday or Sunday flights.

The most important metric in this phase is block time error. It separates the schedule planning question from the broader delay question. A positive average block time error suggests that the scheduled block may be too tight for that route or operating condition. A high late-arrival rate with a negative or near-zero block time error suggests the issue may be driven more by departure delay, taxi-out congestion, late aircraft, or other operational factors.

5. Dataset Overview

Month	Flights	Late Rate (%)	Avg Block Error	Avg Dep Delay	Avg Arr Delay	Avg Taxi Out
October	6483	22.4	-4.4	13.9	9.5	19.7
November	5631	24.7	-5.1	16.1	11.0	20.2
December	5634	25.8	-5.0	17.2	12.2	20.2

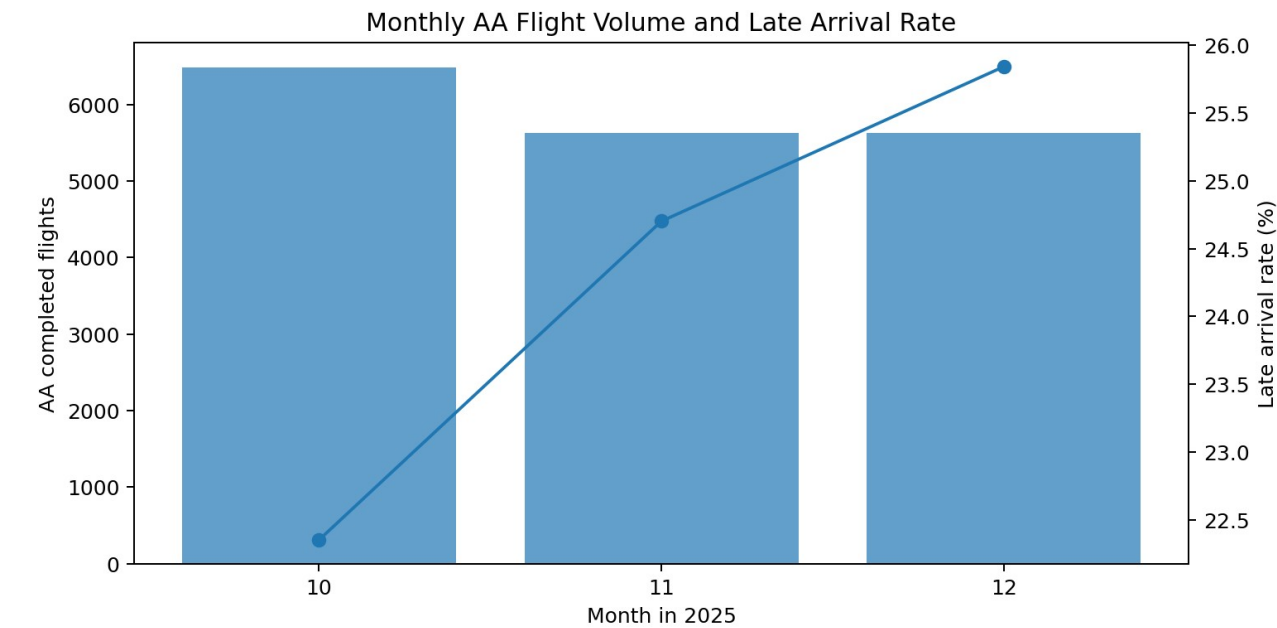


Figure 1. Monthly AA flight volume and late-arrival rate for the cleaned dataset.

The monthly view shows that the number of analyzed AA flights was broadly stable across the three months. Late-arrival rate increased from October to December, moving from 22.4% in October to 25.8% in December. This indicates that the later modeling phase should include month or seasonal indicators rather than pooling all flights without time context.

6. Route-Level Reliability Findings

6.1 Top Late-Arrival Routes

A route-level summary was created using routes with at least 30 flights. This threshold reduces the chance of drawing conclusions from very small samples. The table below shows the highest late-arrival-rate routes among routes meeting that minimum flight threshold.

Route	Flights	Late Rate (%)	Avg Block Error	Avg Dep Delay	Avg Taxi Out	Avg Arr Delay
BDL-DCA	47	42.6	-2.6	40.6	16.8	38.0
DCA-BDL	48	39.6	-5.4	43.9	22.2	38.5
DCA-BOS	626	35.8	0.7	24.4	23.5	25.1
CLT-DCA	595	35.6	4.2	11.3	29.5	15.5
DCA-BNA	89	34.8	0.1	16.9	19.2	17.0
DCA-SYR	49	34.7	-6.1	21.8	23.2	15.7
SYR-DCA	49	34.7	4.0	19.2	21.7	23.3
RIC-DFW	264	32.2	1.0	17.0	17.5	18.0
FLL-DCA	156	32.1	-3.8	19.9	18.9	16.1
STL-DCA	32	31.2	-0.3	15.8	16.7	15.6

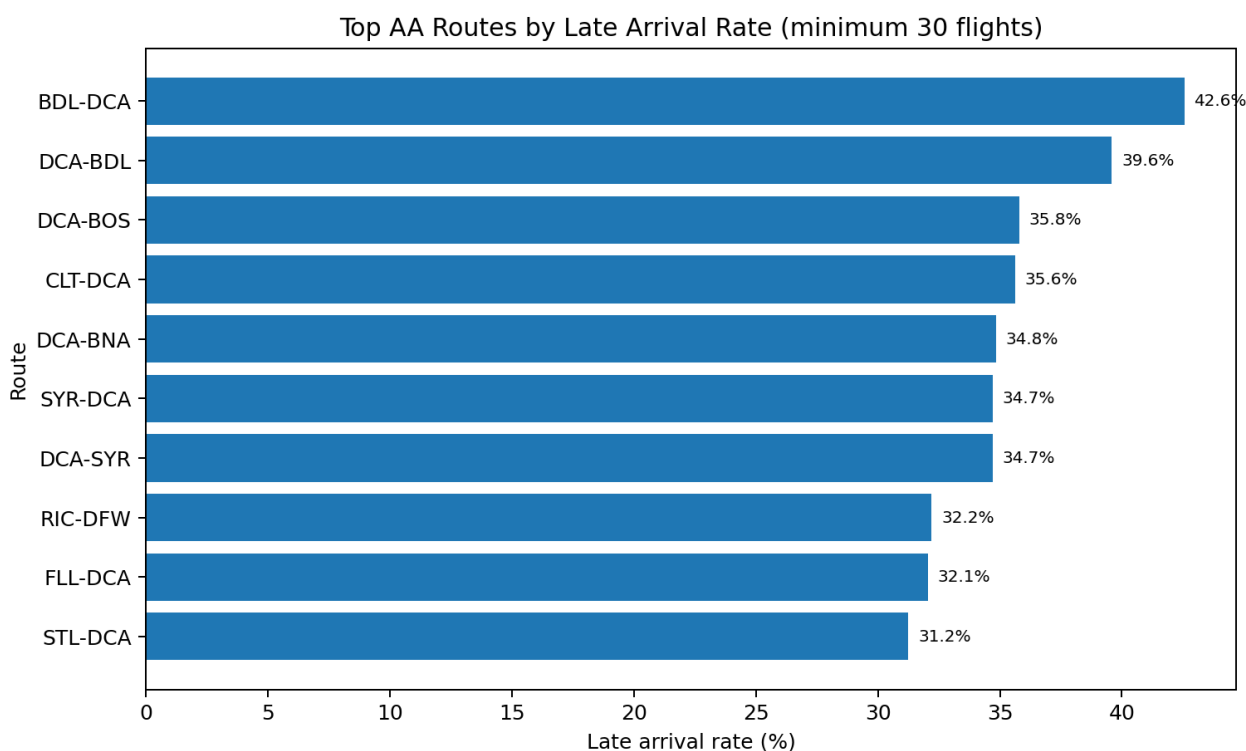


Figure 2. Highest late-arrival-rate AA routes with at least 30 flights.

The top late-arrival routes include BDL-DCA, DCA-BDL, DCA-BOS, CLT-DCA, and DCA-BNA. A key point is that these routes do not all have the same block-time pattern. For example, CLT-DCA has both a high late-arrival rate and a positive average block-time error, while BDL-DCA and DCA-BDL have high late-arrival rates but negative average block-time error. This means route reliability cannot be explained using block time alone.

6.2 Most Underblocked Routes

The next table sorts routes by average block time error. Positive values indicate that actual elapsed time was greater than scheduled elapsed time on average. These routes are candidates for closer block-time review, especially if they also have high late-arrival rates.

Route	Flights	Avg Sched Block	Avg Actual Block	Avg Block Error	Late Rate (%)
CLT-DCA	595	84.2	88.4	4.2	35.6
SYR-DCA	49	85.1	89.2	4.0	34.7
PVD-DCA	113	91.2	92.7	1.5	22.1
RIC-DFW	264	209.6	210.6	1.0	32.2
DCA-BOS	626	95.1	95.7	0.7	35.8
DCA-BNA	89	120.8	120.9	0.1	34.8
RDU-DCA	48	67.8	67.8	0.1	14.6
STL-DCA	32	121.9	121.6	-0.3	31.2
ORD-DCA	534	114.7	113.8	-1.0	28.5
BGR-DCA	62	114.6	113.7	-1.0	17.7

The strongest underblocked candidates in this phase are CLT-DCA and SYR-DCA. Both have average block-time errors above four minutes and late-arrival rates above 34%. These routes are the clearest first-pass candidates for reviewing schedule buffer assumptions.

7. Block-Time Error Distribution

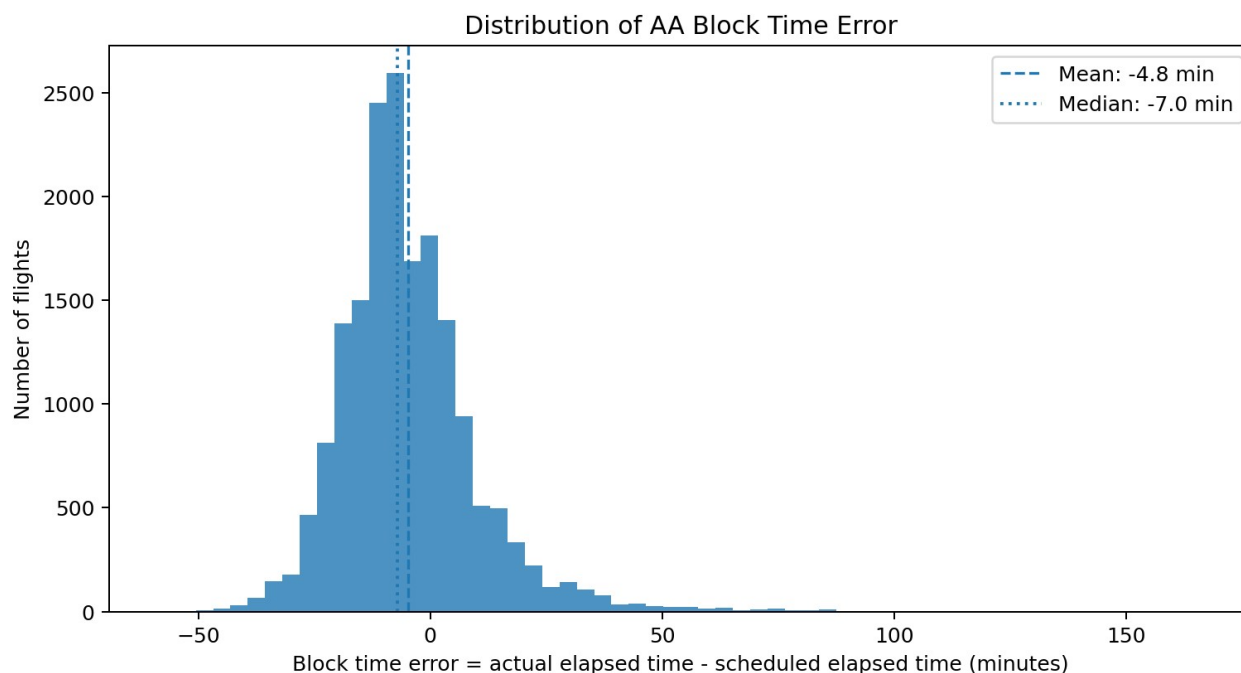


Figure 3. Distribution of individual-flight block time error.

The average block-time error across the cleaned dataset was -4.8 minutes and the median was -7.0 minutes. This means that, on average, actual elapsed time was lower than scheduled elapsed time. However, the distribution still has a right tail: the 75th percentile was 2.0 minutes and the 90th percentile was 12.0 minutes. In practical terms, many flights have enough or excess scheduled block time, but a meaningful subset still exceeds scheduled block time by a non-trivial amount.

This is why the next modeling phase should not simply add the same buffer to every route. A better approach is to predict actual elapsed time by route, departure hour, month, taxi-time patterns, and other relevant operational features.

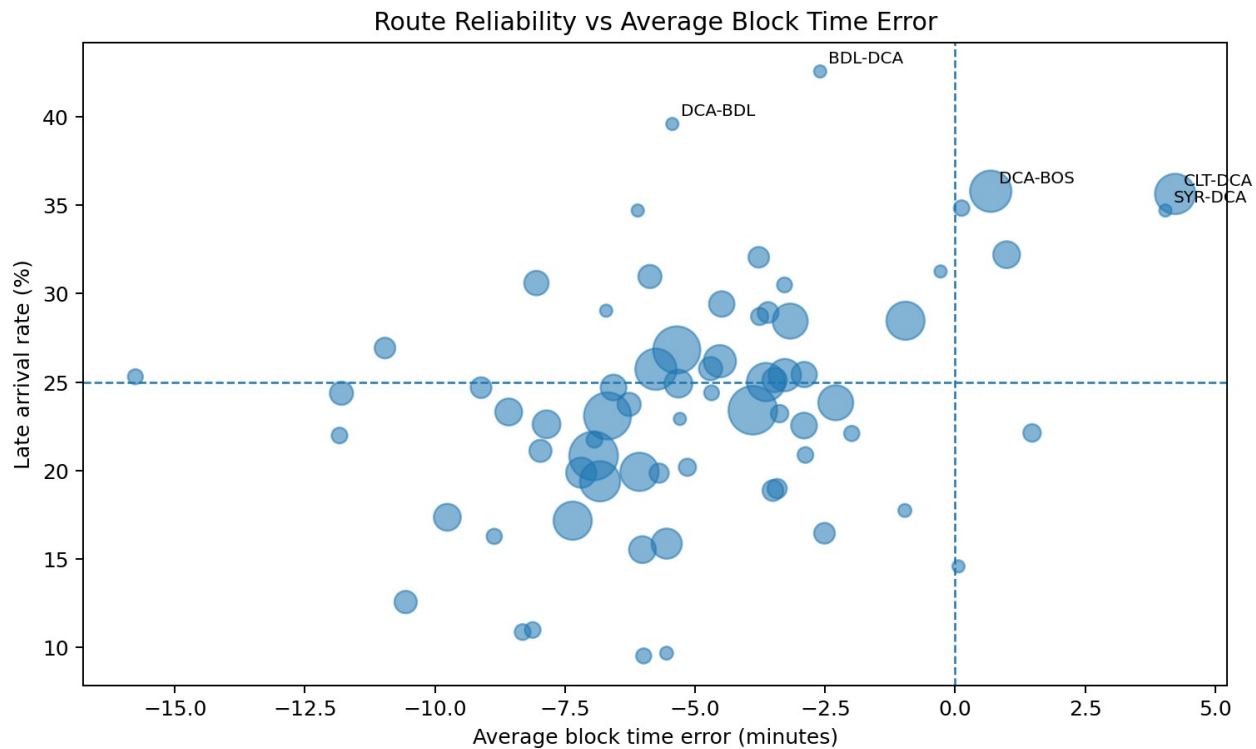


Figure 4. Route-level late-arrival rate compared with average block-time error.

The scatter plot shows that route reliability and block-time error are related but not identical. Routes in the upper-right area have both high late-arrival rate and positive block-time error, making them direct block-time review candidates. Routes in the upper-left area are late despite having negative average block-time error, suggesting the root causes may be upstream or operational rather than purely schedule block-time related.

8. Routes That Are Late but Not Underblocked

This table highlights routes where the late-arrival rate is at least 25%, but the average block-time error is zero or negative. These are important because adding more scheduled block time may not be the first or only solution. The more likely next step is root-cause analysis around departure delay, taxi-out delay, late aircraft, or network congestion.

Route	Flights	Late Rate (%)	Avg Block Error	Avg Dep Delay	Avg Taxi Out	Avg Arr Delay
BDL-DCA	47	42.6	-2.6	40.6	16.8	38.0
DCA-BDL	48	39.6	-5.4	43.9	22.2	38.5
DCA-SYR	49	34.7	-6.1	21.8	23.2	15.7
FLL-DCA	156	32.1	-3.8	19.9	18.9	16.1
STL-DCA	32	31.2	-0.3	15.8	16.7	15.6
DFW-IAD	197	31.0	-5.9	20.2	18.8	14.3
DFW-ORF	219	30.6	-8.1	23.7	17.5	15.7
SRQ-DCA	82	30.5	-3.3	27.0	19.6	23.8

For example, BDL-DCA and DCA-BDL both had late-arrival rates above 39%, but negative average block-time error. Their average departure delays were over 40 minutes, which points toward pre-departure or upstream reliability issues rather than simply a lack of scheduled gate-to-gate time.

9. Departure-Hour Reliability Patterns

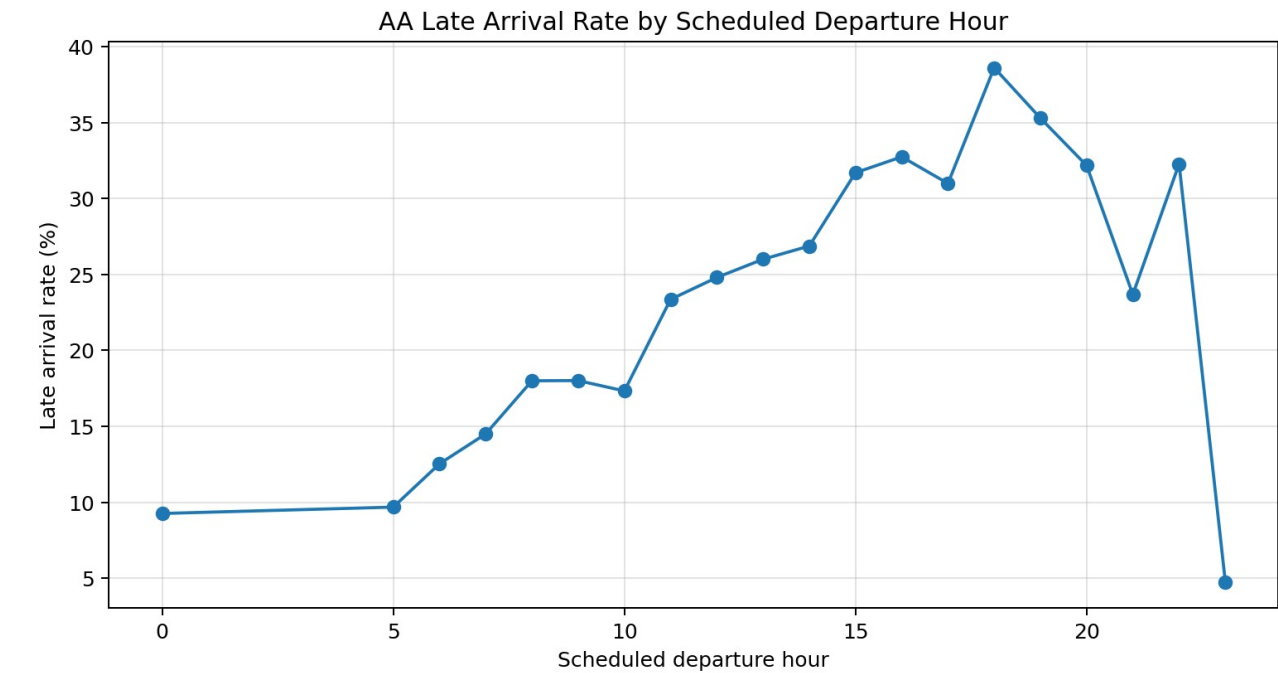


Figure 5. Late-arrival rate by scheduled departure hour.

Dep Hour	Flights	Late Rate (%)	Avg Block Error	Avg Dep Delay	Avg Taxi Out
18	1049	38.6	-2.6	25.1	20.4
19	972	35.3	-5.7	26.6	19.9
16	507	32.7	-3.6	23.3	19.6
22	409	32.3	-5.7	19.8	19.2
20	830	32.2	-4.3	21.7	20.9
15	1161	31.7	-3.3	20.8	20.6
17	1478	31.0	-3.8	20.5	20.8
14	1057	26.9	-4.3	18.5	19.3

The late-arrival rate is highest in the late afternoon and evening departure windows. The 18:00 hour had the highest late-arrival rate at 38.6%. The high-risk hours also show elevated average departure delay, which is consistent with operational delay accumulation through the day. This finding supports adding departure-hour features in the modeling phase.

10. Initial Schedule Buffer Recommendation Framework

A rule-based recommendation table was created as a first decision-support output. The purpose of this table is not to finalize schedule changes. Instead, it converts the EDA into an actionable routing framework for deeper review.

Category	Rule
Increase block time / review schedule buffer	Average block-time error greater than 3 minutes and late-arrival rate greater than 25%.
Potentially reduce excess block time	Average block-time error less than -5 minutes and late-arrival rate less than 20%.
Investigate operational root causes	Late-arrival rate greater than 25% and average block-time error less than or equal to 0 minutes.
Monitor	Routes not meeting the above rule thresholds.

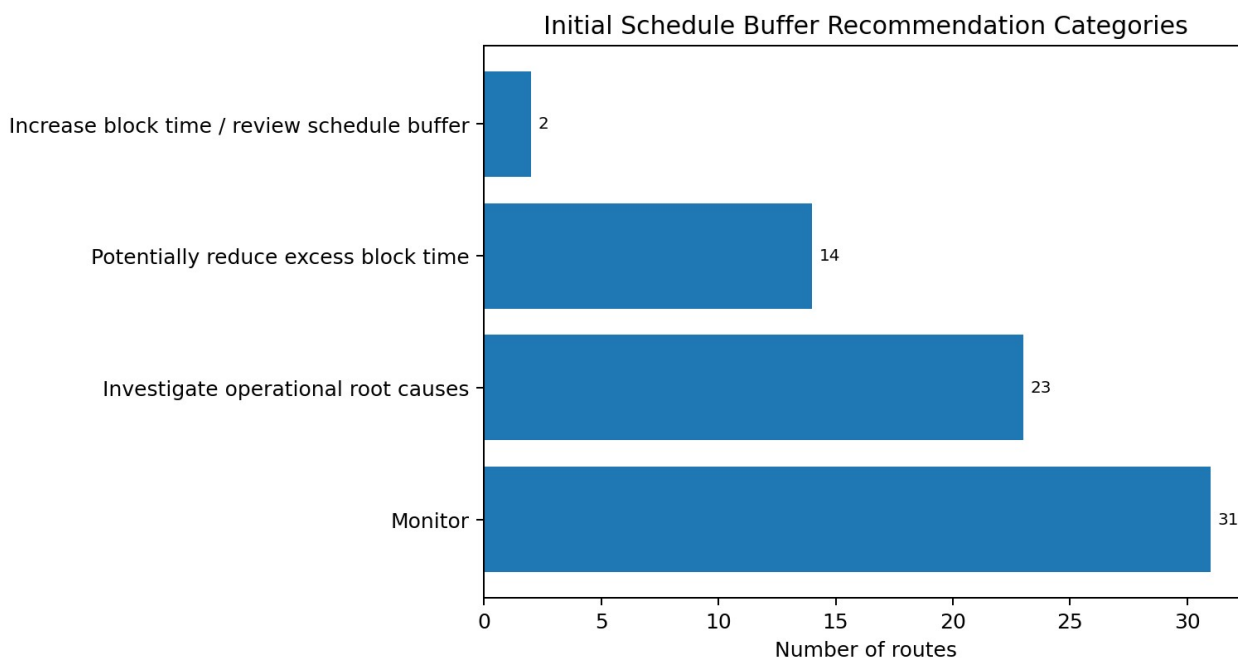


Figure 6. Number of routes by initial recommendation category.

Recommendation Category	Number of Routes
Monitor	31
Investigate operational root causes	23
Potentially reduce excess block time	14
Increase block time / review schedule buffer	2

The initial rules identified two routes for block-time buffer review, twenty-three routes for operational root-cause investigation, fourteen routes as possible excess-block candidates, and thirty-one routes to monitor. This split is useful because it prevents the analysis from treating every late route as a block-time problem.

10.1 Routes Recommended for Block-Time Review

Route	Flights	Late Rate (%)	Avg Block Error	Avg Sched Block	Avg Actual Block	Avg Dep Delay	Avg Taxi Out
CLT-DCA	595	35.6	4.2	84.2	88.4	11.3	29.5
SYR-DCA	49	34.7	4.0	85.1	89.2	19.2	21.7

10.2 Selected Potential Excess-Block Candidates

Route	Flights	Late Rate (%)	Avg Block Error	Avg Sched Block	Avg Actual Block
LAX-DCA	183	12.6	-10.6	294.6	284.1
DCA-PHX	265	17.4	-9.8	312.5	302.7
DCA-VPS	86	16.3	-8.9	161.5	152.6
ORF-MIA	92	10.9	-8.3	146.2	137.9
DCA-LAS	91	11.0	-8.1	320.3	312.2
RIC-CLT	536	17.2	-7.4	86.1	78.8
CLT-ORF	332	19.9	-7.2	75.0	67.8
DCA-CLT	593	19.4	-6.8	96.8	89.9
CLT-RIC	537	19.9	-6.1	73.7	67.6
PHX-DCA	264	15.5	-6.0	251.9	245.9

11. Key Findings From Part 1

1. The cleaned dataset contains 17,748 completed AA flights across 91 routes, which is sufficient for initial route-level EDA and a first modeling prototype.
2. The overall late-arrival rate was 24.2%, with reliability becoming worse from October to December in this dataset.
3. The average block-time error was -4.8 minutes, so the system is not uniformly underblocked. Some routes may even have excess scheduled block time.
4. High late-arrival rate does not automatically imply insufficient scheduled block time. Several late routes had negative average block-time error, suggesting operational root causes outside the gate-to-gate block-time estimate.
5. CLT-DCA and SYR-DCA are the clearest first-pass candidates for block-time buffer review because both have high late-arrival rates and positive average block-time error.
6. Late afternoon and evening flights showed higher late-arrival rates, supporting the inclusion of departure hour in the modeling phase.
7. The recommendation framework provides an initial triage structure: review block buffers, investigate operational causes, consider excess block, or monitor.

12. Limitations of the First Phase

- The analysis is descriptive. It identifies patterns and candidates for deeper review but does not prove causality.
- Only three months of data were used. A full-year dataset would better capture seasonality, weather patterns, and route stability.
- The project focuses only on AA flights connected to the selected Virginia/DC-area airports, so the conclusions should not be generalized to the entire airline network.

- Delay cause fields are mainly populated for delayed flights and may not fully explain all reliability issues.
- The recommendation rules are intentionally simple. They should later be replaced or refined using prediction models and scenario analysis.

13. Recommended Next Steps

1. Build a regression model to predict actual elapsed time using route, month, day of week, departure hour, scheduled elapsed time, distance, taxi times, and other operational features.
2. Build a classification model to predict late-arrival risk using the same features plus delay-related variables available before or near departure.
3. Compare baseline models such as linear regression and logistic regression with tree-based models such as random forest or gradient boosting.
4. Create route-level predicted percentile block times, such as 70th, 80th, or 90th percentile recommendations, instead of relying only on average block-time error.
5. Use the model outputs to evaluate block strategies: lean, balanced, and conservative schedules.
6. Create a dashboard summarizing route reliability, block-time variance, underblocked routes, late-but-not-underblocked routes, and recommendation categories.

Appendix A. Columns Used in the First Phase

Column	Use in Analysis
crs_elapsed_time	Scheduled elapsed/block time in minutes.
actual_elapsed_time	Actual gate-to-gate elapsed time in minutes.
arr_delay	Arrival delay in minutes.
dep_delay	Departure delay in minutes.
taxi_out	Taxi-out time in minutes.
taxi_in	Taxi-in time in minutes.
distance	Flight distance in miles.
origin, dest	Route endpoints.
crs_dep_time	Scheduled departure time used to derive departure hour.
cancelled, diverted	Used to remove cancelled and diverted records from block-time modeling data.